

NE235 – Nuclear Reactor Operations Training

Lab Module #0 – PULSTAR Reactor Facility Design & Specifications

1. Mission and Applications
2. Plant Layout & Experimental Facilities
3. Reactor Fuel & Core Specifications
4. Reactor Systems
 - Reactor Control & SCRAM Logic
 - Primary & Secondary Cooling Loops
 - HVAC & Confinement
 - Radiation Monitoring & Evacuation

PULSTAR Reactor – Mission & Applications

Mission:

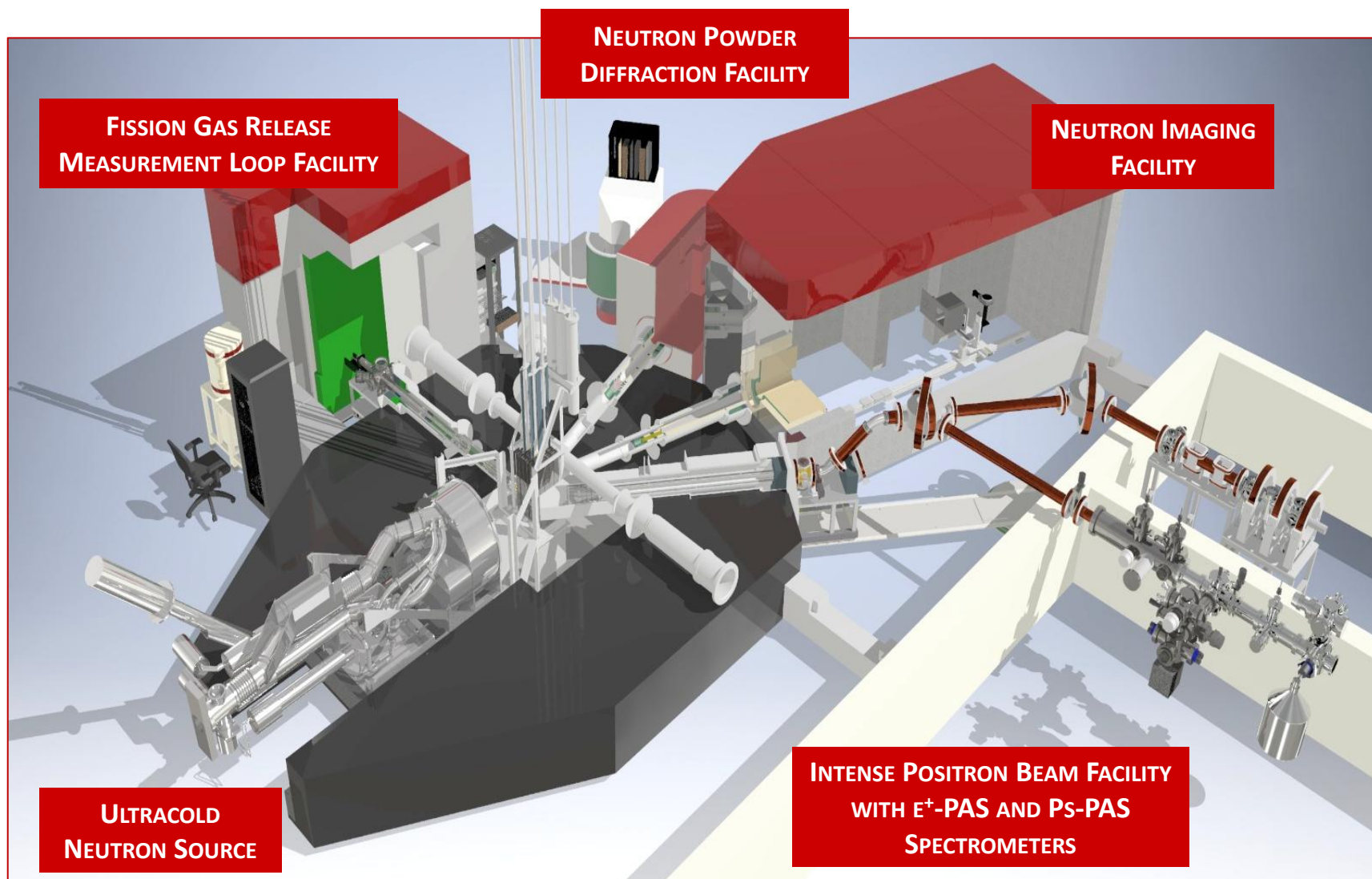
- Education, Research, Training, Service

Applications:

- Academic Laboratories
- Reactor Operator Training
- Distance Learning – Teleconferencing Labs
- Materials Research – E⁺ Beam (IPB) with E+PALS/DBS spectrometers, Neutron Diffractometer (NDF), Fission Gas Release Measurement Loop (FGRML), Extreme Environment Irradiation Facility (EEIF)
- Fundamental Physics – UCN source
- Trace Element Analysis – environmental, forensic, industrial
- Non-Destructive Testing – neutron imaging (NIF)
- Detector Calibration – Navy, Power reactors
- Isotope Production – ¹⁶⁶Ho; ⁹⁹Mo



PULSTAR Reactor – Experimental Facilities



New Facilities & Upgrades

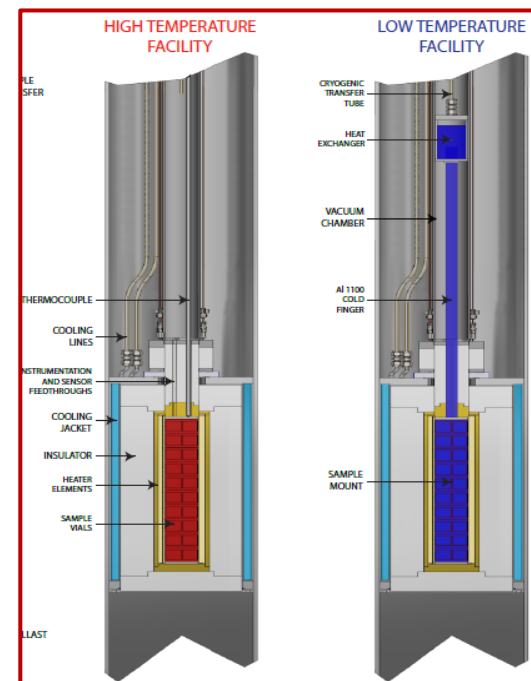


New Facilities:

- **Hot Cell** – *isolated environment with 6" lead shielding and mating pool transfer cask*
- **Extreme Environment Irradiation Facility (EEIF):**
+800°C for advanced reactor R&D (sensors, MSR materials).

Upgrades:

- **2 MW Power Upgrade:**
license approval pending with NRC
- **E+ Beam PALS/DBS upgrades:** *Temperature stages; increased spatial and timing resolution; coincidence DBS.*



PULSTAR – Fuel Assembly

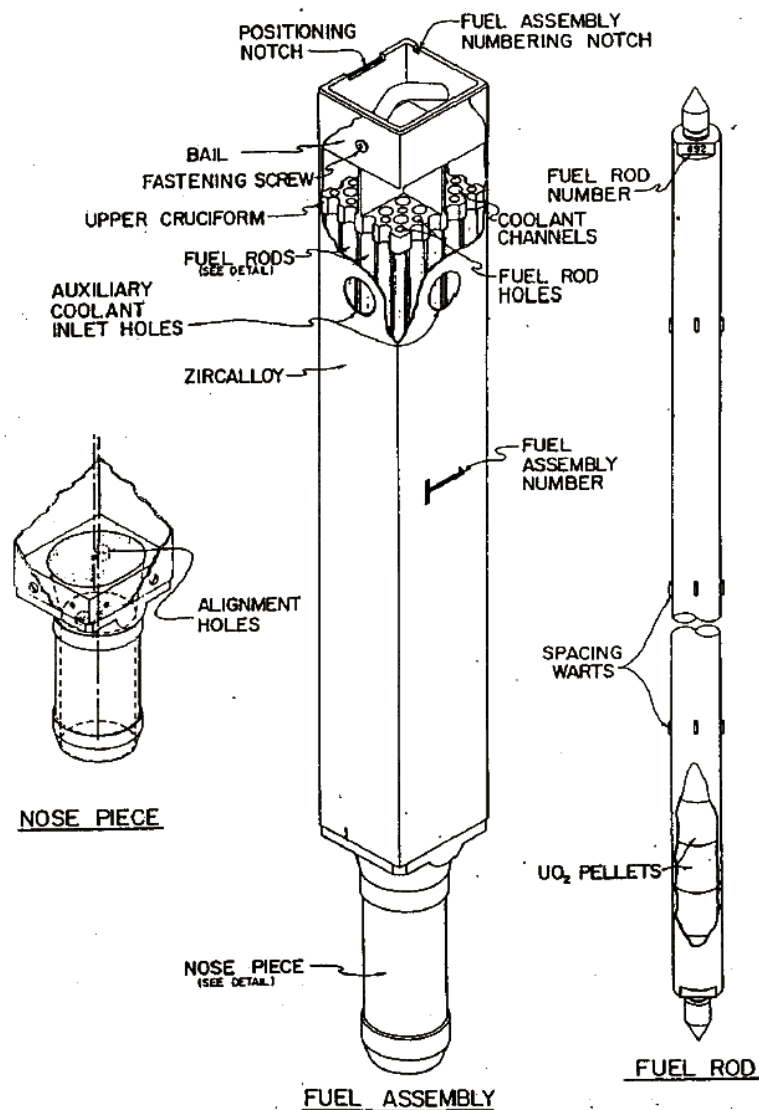


Figure 4.3 PULSTAR fuel

Fuel

Material
 Form
 Enrichment (weight % U-235)
 Design Inventory Core (kg UO₂)
 Density (gm/cm³)
 U-235 per Fuel Pin (gm)

UO₂
 Sintered Pellets
 4% + 6%
 359
 10.5 to 10.76
 20.02

Fuel Pin

Pellet (diameter in (cm))
 Diametrical Gap (in (cm))
 Zircaloy-2 Clad Thickness (in (cm))
 Outside Diameter Pin (in (cm))
 Rectangular Spacing (center-to-center, in (cm))
 Clearance (pin-to-pin, in (cm))
 Clearance (pin-to-box, in (cm))
 Height of Pellet Stack (in (cm))
 Pins per Core
 Height of Pellet (in (cm))

0.423 (1.074)(nominal)
 0.0085 (0.0216)(nominal)
 0.0205 (0.0521)(nominal)
 0.4725 (1.2002)(nominal)
 0.606 x 0.524 (1.54 x 1.33)
 0.051 x 0.133 (0.130 x 0.338)
 0.025 x 0.066 (0.064 x 0.168)
 24 (61)
 625
 0.60 (1.52)

Fuel Box

Material
 Inside Dimensions (in (cm))
 Wall Thickness (in (cm))
 Clearance between Assemblies (in (cm))
 Clearance between Control Rod Guide and Assemblies (in (cm))
 Fuel Pins per Assembly
 Weight (pounds (kg))

Zr-2
 2.620 x 3.030 (6.655 x 7.700)
 0.060 (0.152)
 0.040 (0.102)
 0.060 (0.152)
 25
 44 (20)

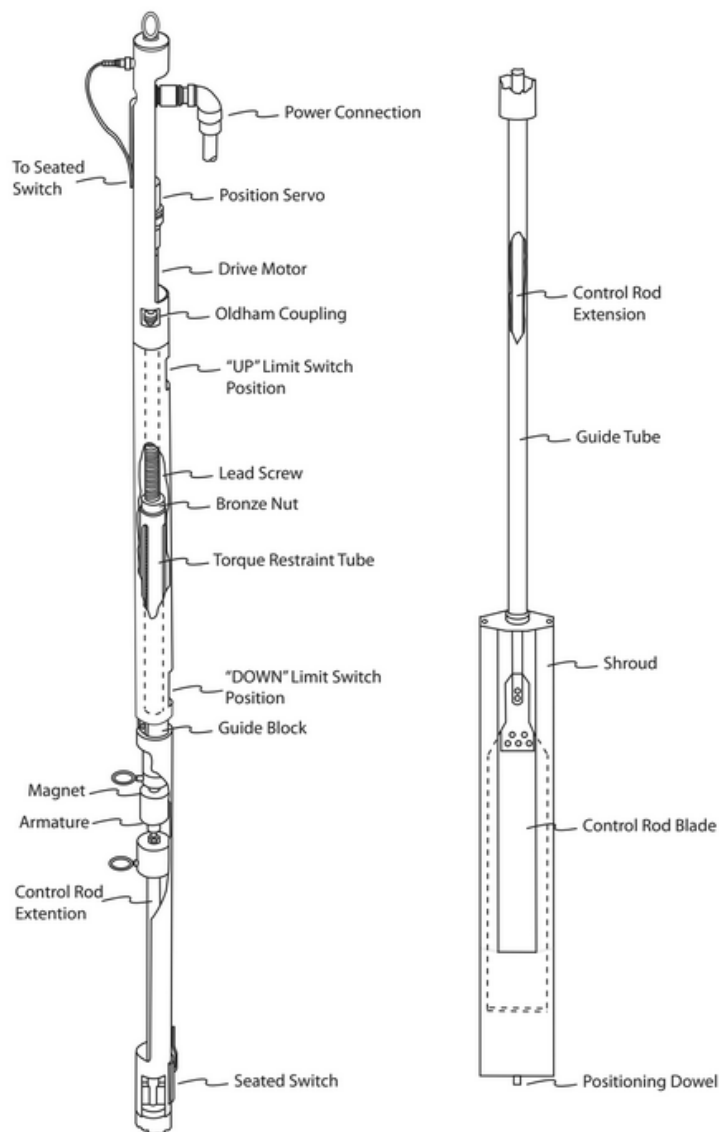
Training Tip: Many of these will need to be memorized!

Reactor Core Specifications & Reactivity Coefficients

Core Dimensions

Overall (in (cm))	15 $\frac{7}{8}$ x 15 (40.3 x 38)	Minimum Critical Assemblies	20
Height (in (cm))	24 (61)	Doppler Coefficient (pcm/°F)	-1.40
Volume Fractions		Void Coefficient (pcm/cm ³)	-1.03
UO ₂	0.3823	Moderator Temperature Coefficient (pcm/°F)	-3.2
Gap (Helium)	0.0155	Beam Tube Worths (air to water- filled pcm)	
Cladding (includes warts)	0.0803	6 in (15.2 cm) diameter	193
Lattice Water	0.3858	8 in (20.3 cm) diameter	45
Assemblies (Zr-2)	0.0753	12 in x 12 in (30.5 cm x 30.5 cm)	20
Water Between Assemblies	0.0255		
Control Rod Guides (Al)	0.0128		
Water Inside Guides (rods out)	0.0226		
Total Volume Fractions		<u>Moderator, Reflector, and Coolant</u>	
UO ₂	0.3823	Material	Light Water
Helium Gap	0.0155	Nominal Inlet Temperature (°F (°C))	105 (40.6)
H ₂ O	0.4339	Nominal Outlet Temperature (°F (°C))	118.8 (48.2)
Zr-2	0.1555	Primary Coolant Flow Rate (gpm (lps))	500 (31.5)
Aluminum	0.0128	Secondary Coolant Flow Rate (gpm (lps))	700 (44.2)
Core Volume Ratios			
H ₂ /UO ₂	1.135		
H ₂ O/U	2.06		

Control Rod Assembly



Control Rods

Absorber Material

Ag-In-Cd (80%-15%-5%)

Guide Material

Aluminum

Shape

Rectangular

Transverse Dimensions (in (cm))

Guide

6.30 x 0.43 (16.0 x 1.09)

Absorber

4.85 x 0.18 (12.32 x 0.46)

Clearance Absorber to Guide
(in (cm))

0.00625 (0.01588)

Clad Material

Sn/Ni

Number of Control Rods

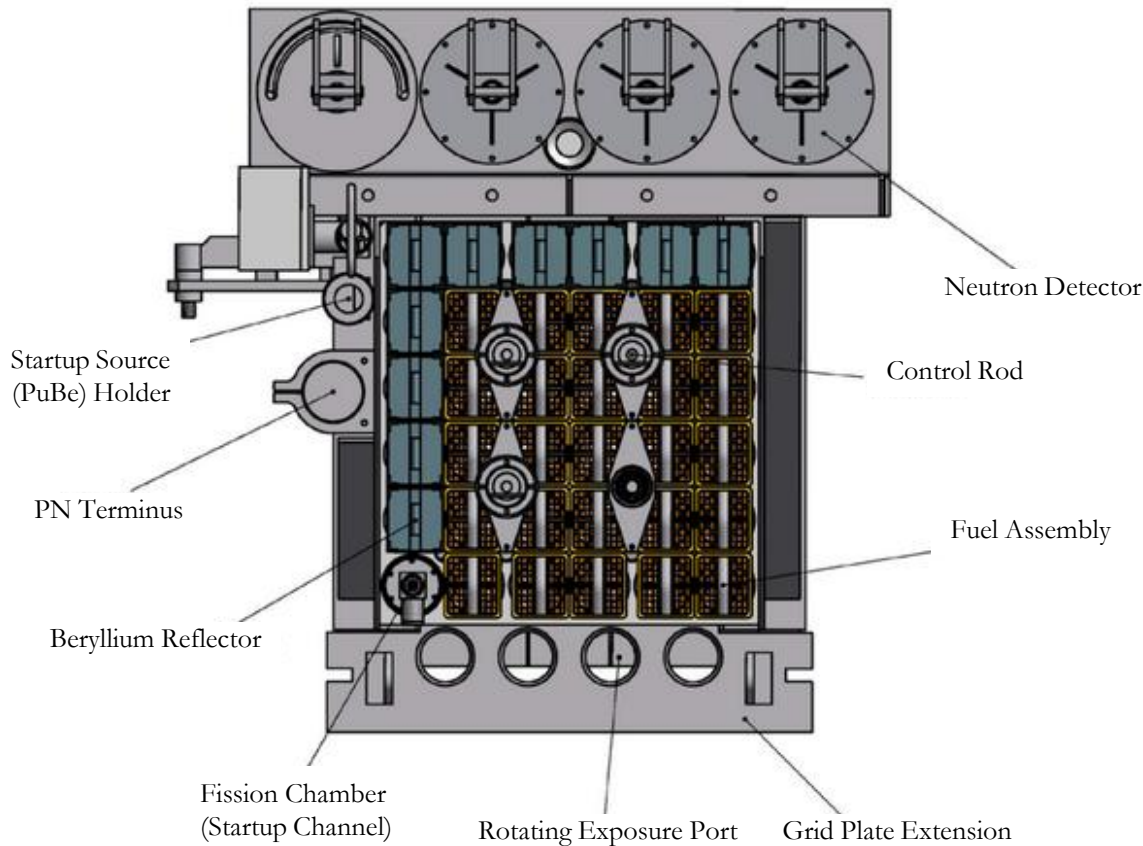
3

Number of Non-Scrammable Rods

1

Training Tip: many of these will need to be memorized!

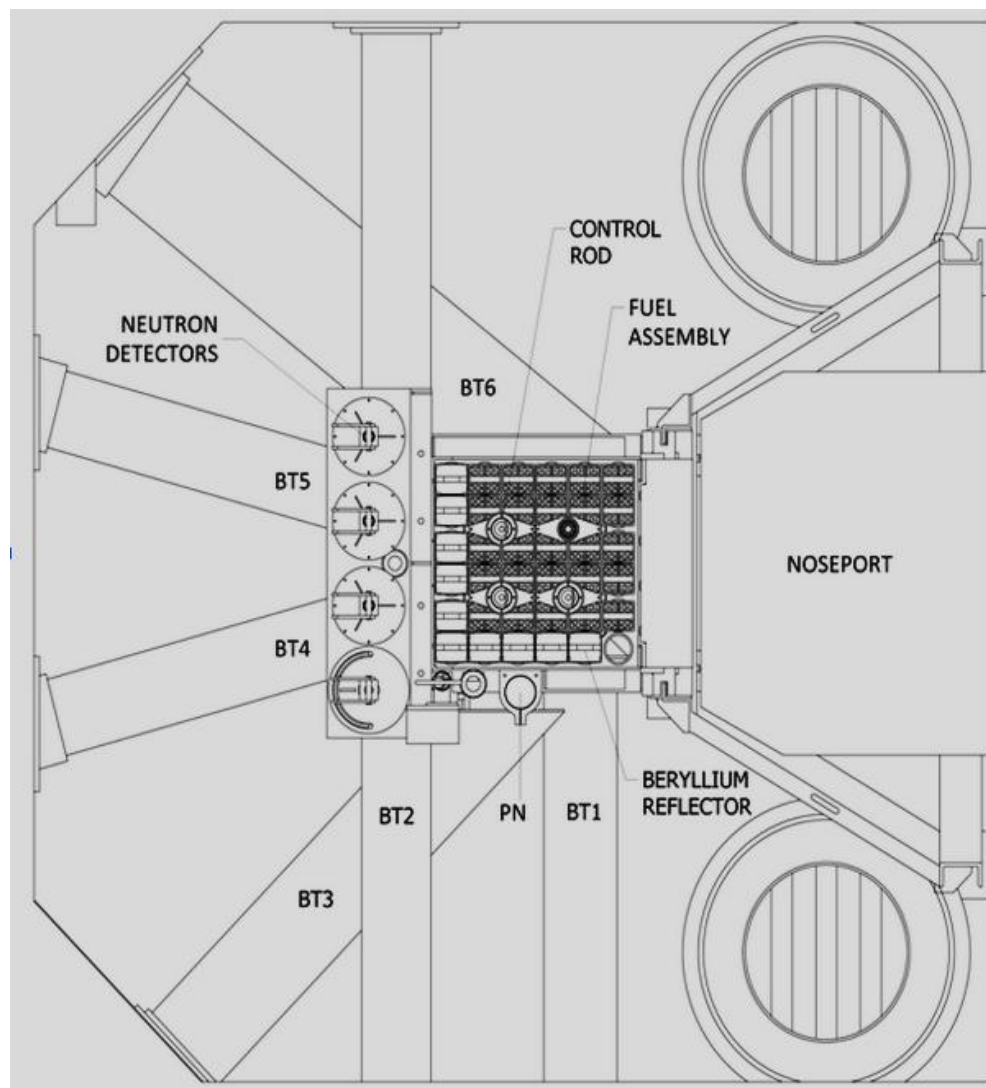
PULSTAR – Core and Gridplate Assembly



Core Structures:

- 36 Position Grid Plate
- 25 Fuel Assemblies
- 10 Reflector Elements
- 4 Control Rods
- 3 NI Power Monitors
- Cooling Plenum
- Flapper Valve Assembly
- Startup Source
- Fission Chamber

In-Pool Irradiation and Beam Port Facilities



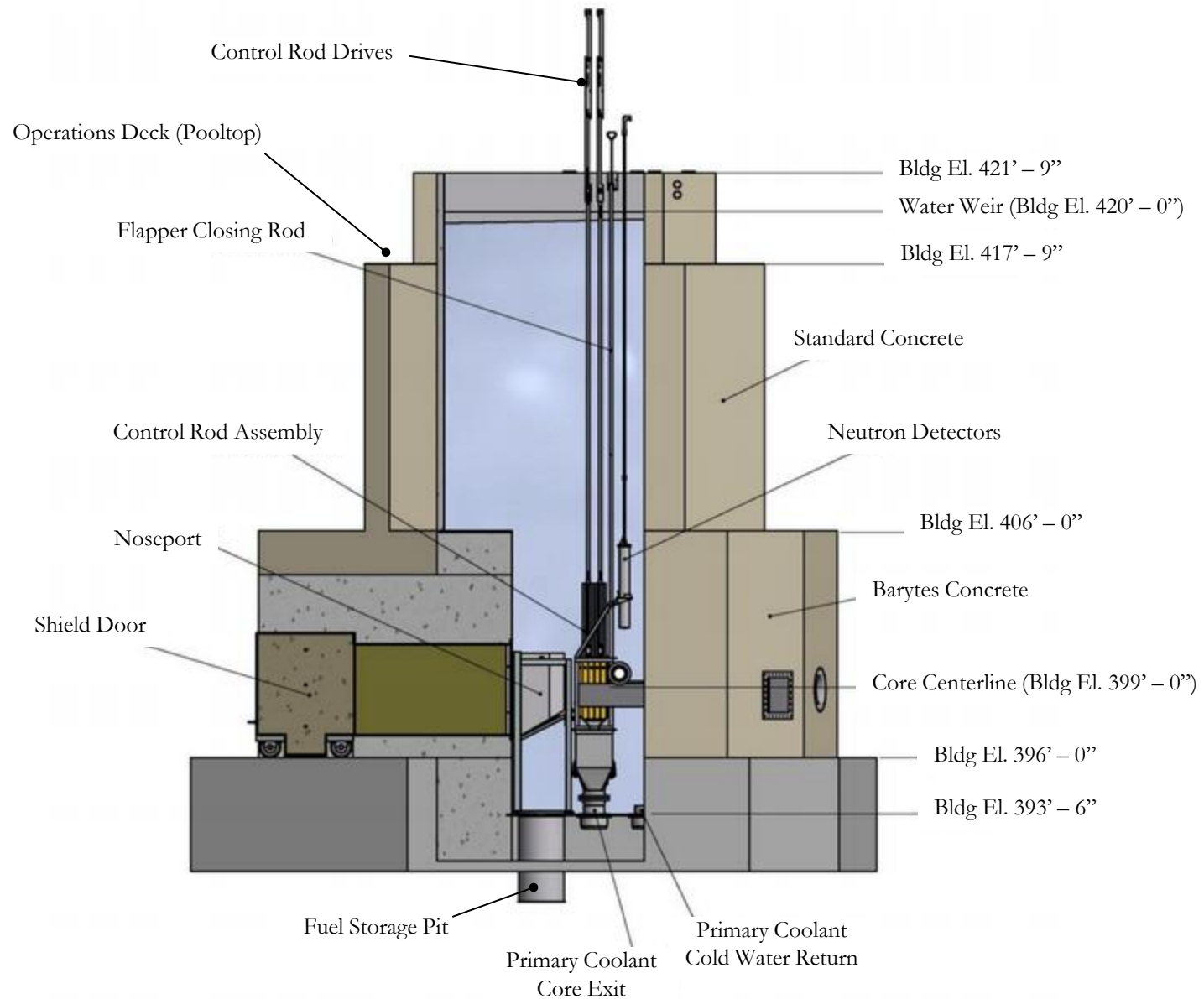
Beam-Port Facilities:

- BP#1 – Fission Gas Measurement Loop Facility
- BP#4 – Neutron Diffractometer
- BP#5 – Neutron Imaging Facility
- BP#6 – Positron Intense Beam Fac.
- Noseport – UCN Source

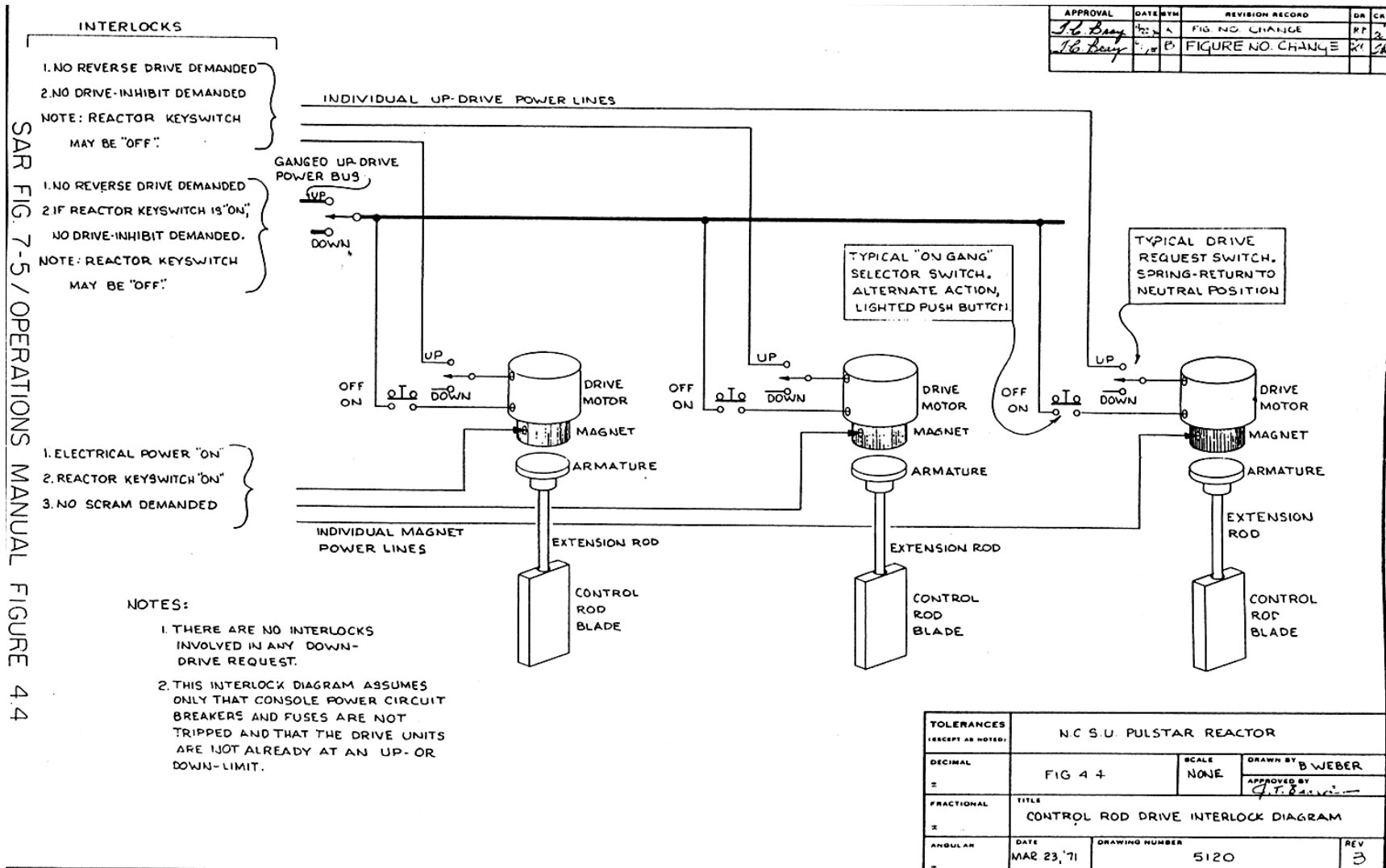
In-Pool Irradiation Facilities:

- 4 Rotating Exposure Ports
- 2 Dry Tubes
- 1 PN Terminus (ex-core)
- 3.5"-8" I.D. Standpipes

Reactor Plant – Elevation View



Control Rod – Electromagnet Assembly



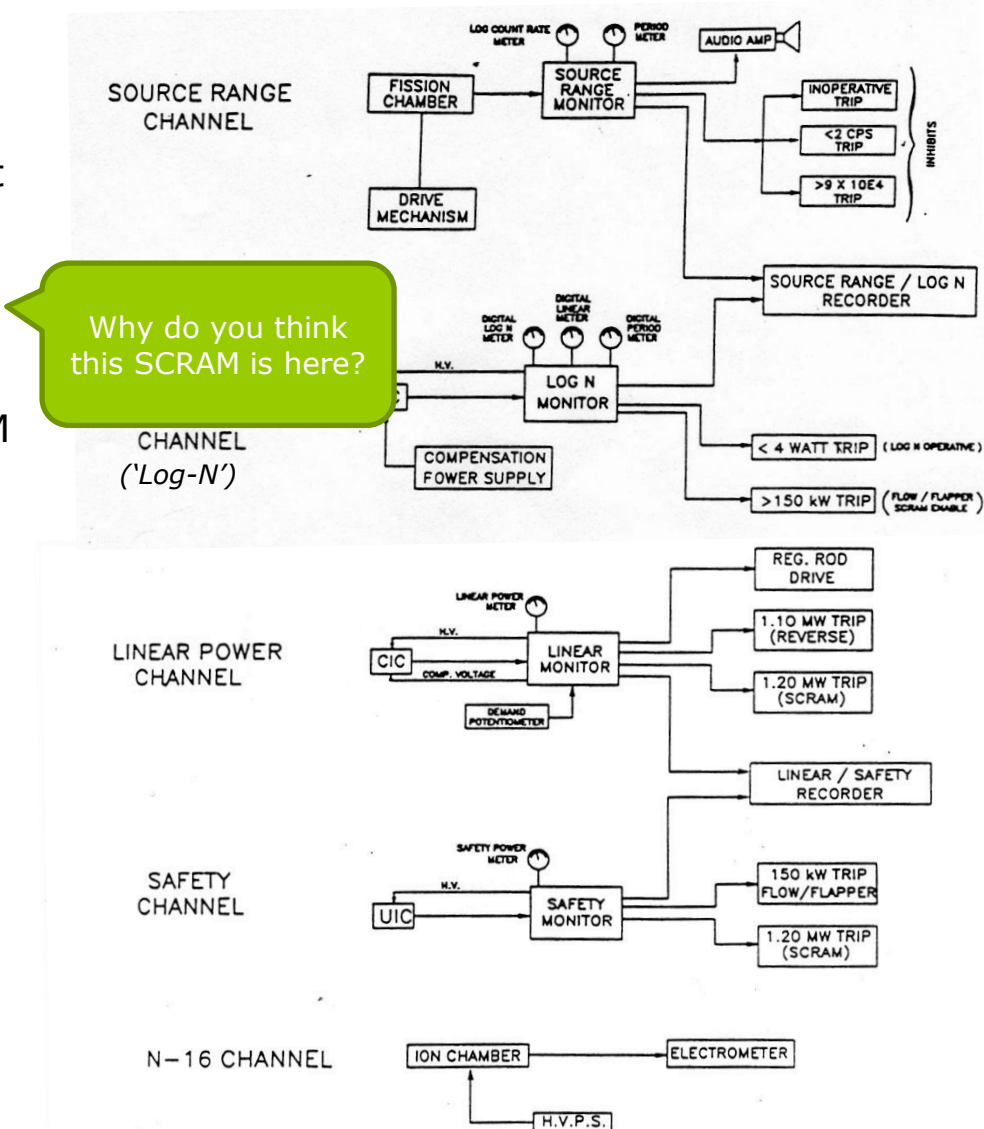
Reactor Instrumentation & SCRAM System

Seven Reactor SCRAM's:

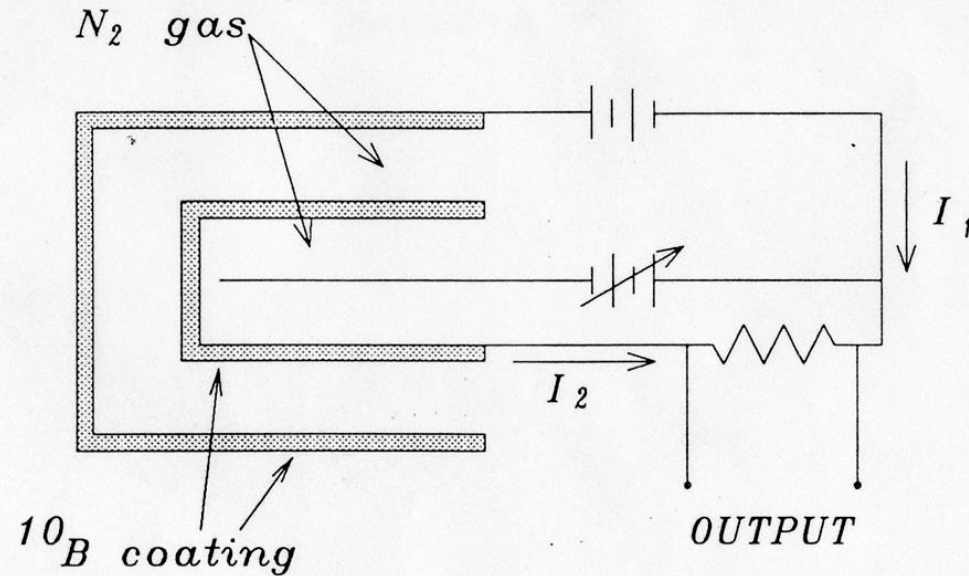
1. Linear Power Channel SCRAM at $\geq 120\%$ Full Power (1MW)
2. Safety Power Channel SCRAM at $\geq 120\%$ Full Power (1MW)
3. Primary Coolant Flow SCRAM at ≤ 475 GPM (*enables if Log-N or Safety Channel ≥ 150 kW*)
4. Flow Flapper Valve OPEN SCRAM (*enables if Log-N or Safety Channel ≥ 150 kW*)
5. Pool Level SCRAM at $\leq 36''$ below full
6. Pool Temperature SCRAM if T_2 (rtd) $\geq 116^\circ\text{F}$
7. Manual SCRAM – Depress Red Button on main console

Plus:

- Console Keyswitch
- SCRAM Logic Ground Fault



Nuclear Instrumentation – CIC Neutron Detector



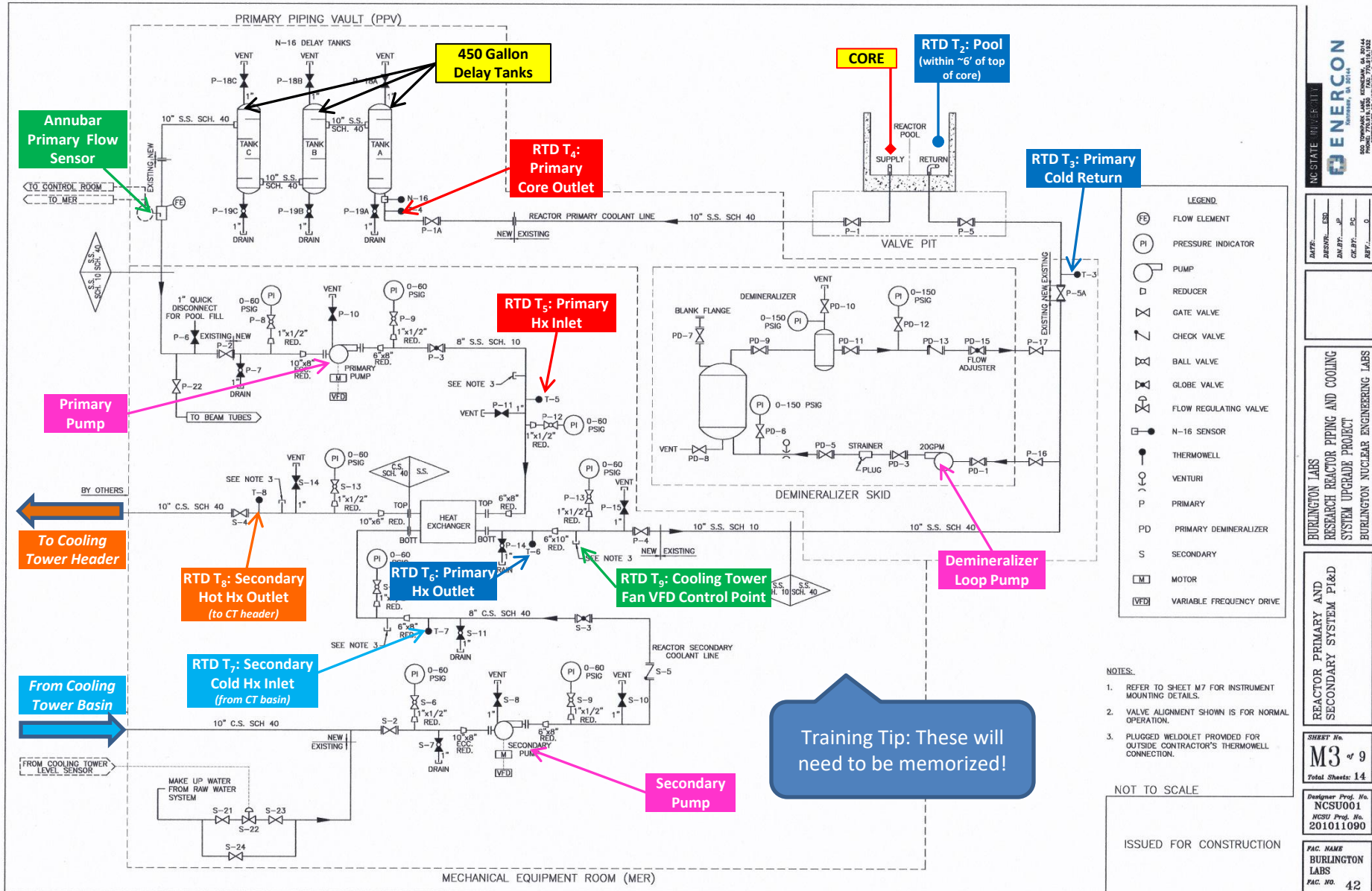
$I_1 =$ Neutron and gamma current

$I_2 =$ Gamma current

$$\text{OUTPUT CURRENT} = I_1 - I_2 = \text{Neutron current}$$

Figure 2.3 Compensated Ion Chamber

PULSTAR - Primary Cooling System Schematic



NO STATE PROJECT
ENERCON
Burlington, VA 20144
800 DOWNEY LANE, GREENSBORO, NC 27404
PROVIDING THE BEST OF CONSTRUCTION

DATE: _____
DESIGN: _____
DRAWN: _____
CHECK: _____
BY: _____

BURLINGTON LABS
RESEARCH REACTOR PIPING AND COOLING
SYSTEM UPGRADE PROJECT
BURLINGTON NUCLEAR ENGINEERING LABS

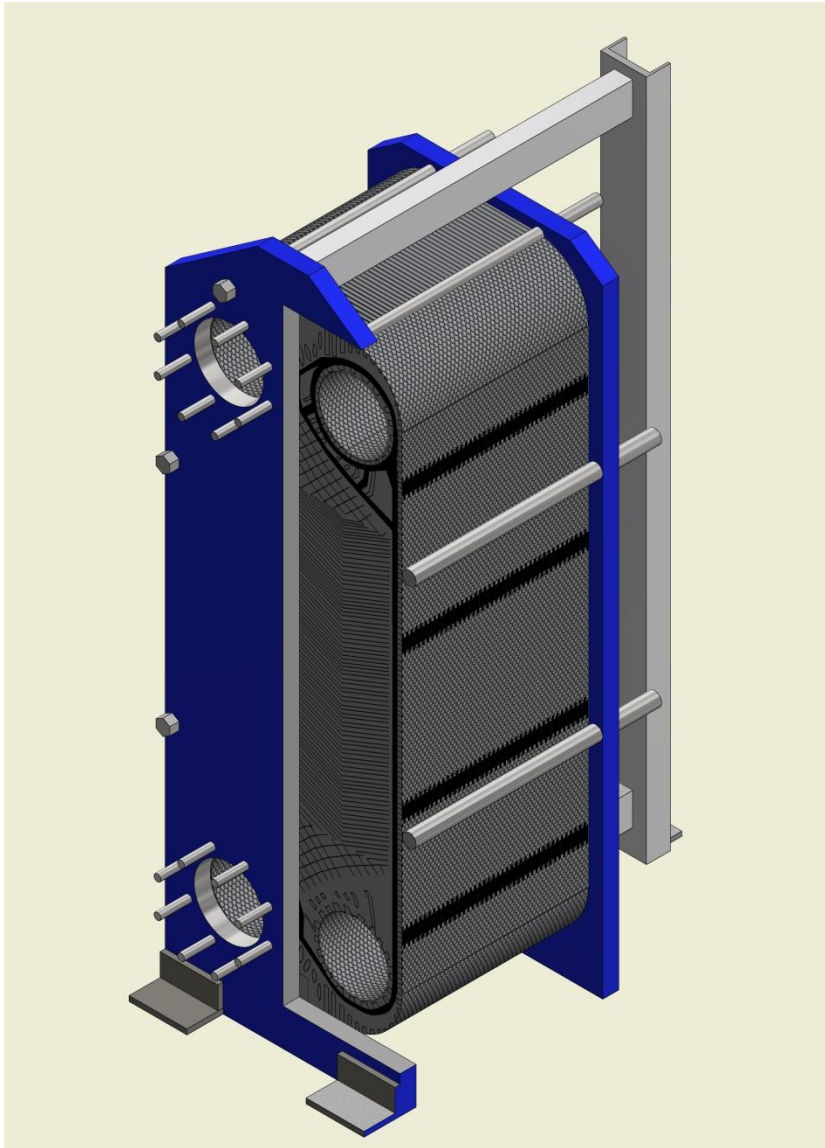
**REACTOR PRIMARY AND
SECONDARY SYSTEM P&ID**

SHEET No.
M3 of 9
Total Sheets: 14

Designer Proj. No.
NCSU001
NCSU Proj. No.
201011090

PAC. NAME
BURLINGTON
LABS
PAC. NO. 42

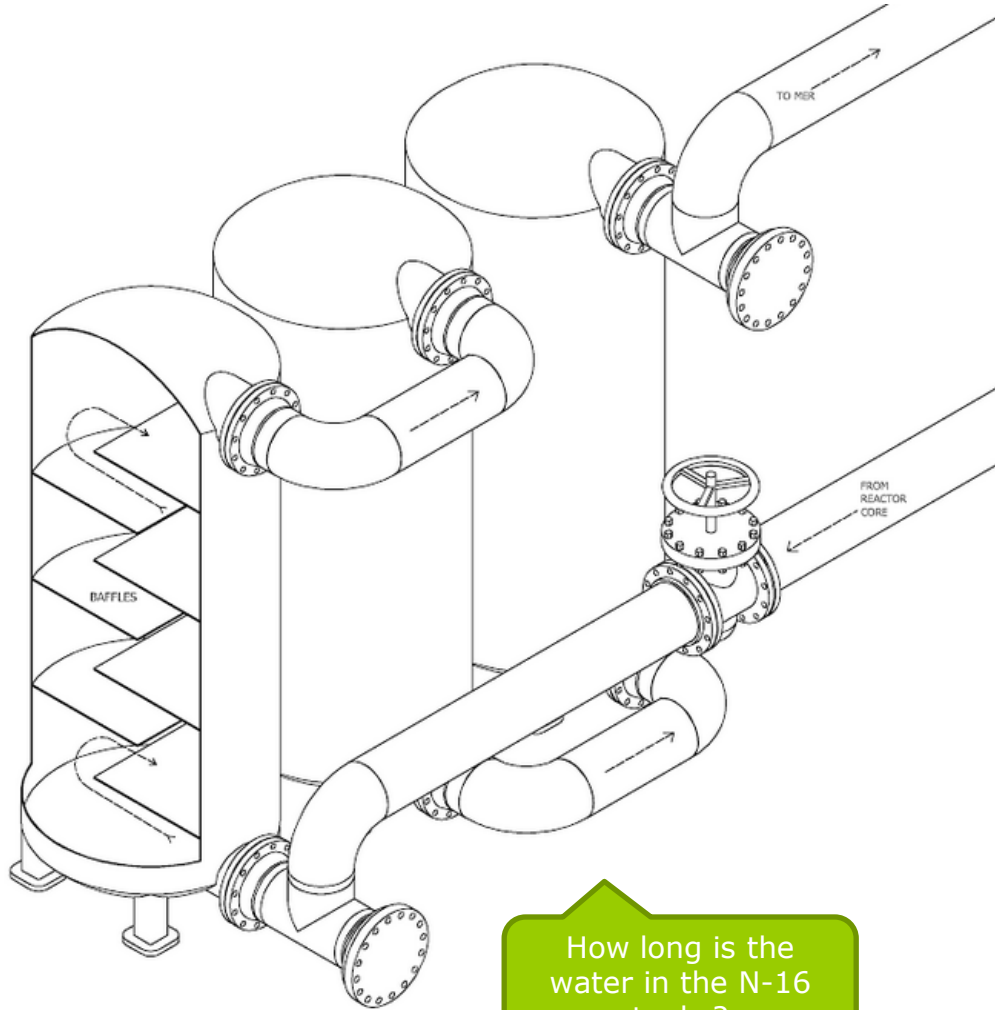
Alfa Laval Plate and Frame Heat Exchanger



Heat Exchanger:

- 126 Plates – 0.5mm thick, 304 Stainless Steel
- 500 GPM Primary Flow Rate
- 700 GPM Secondary Flow Rate
- 2 MW Current Capacity; expandable.
- Design Pressure – 150 PSI
- Max Temp – 200°F
- 4.74 PSI drop at 1000 GPM

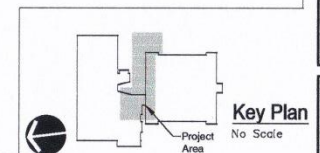
Nitrogen-16 Delay Tanks



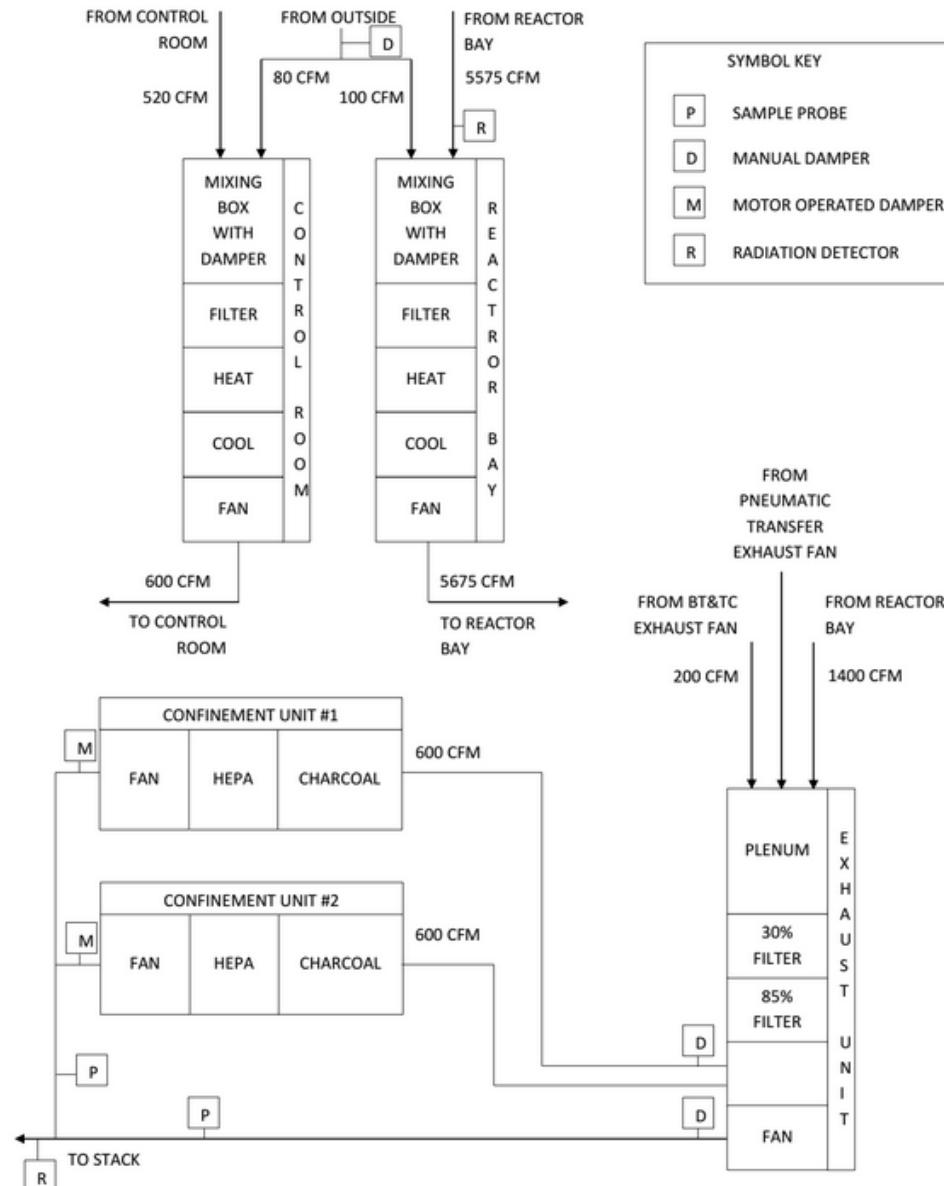
How long is the water in the N-16 tanks?



NC STATE UNIVERSITY DATE: 2/12/2015 DESIGNER: MAH DRAWN BY: MAH CHECKED BY: MAH DATE: NC STATE UNIVERSITY BURLINGTON LABS NUCLEAR REACTOR COOLING COOLING P&ID SHEET No. H3.1 Total Sheets: Designer Proj. No. 12-027 NCST Proj. No. 201011090	EDMONDSON ENGINEERS 10000 N. WILKESBORO RD. SUITE 200 WILKESBORO, NC 27159 TEL: 703.722.8888 FAX: 703.722.8889 FAC. NAME BURLINGTON LABS FAC. NO. 042
--	--

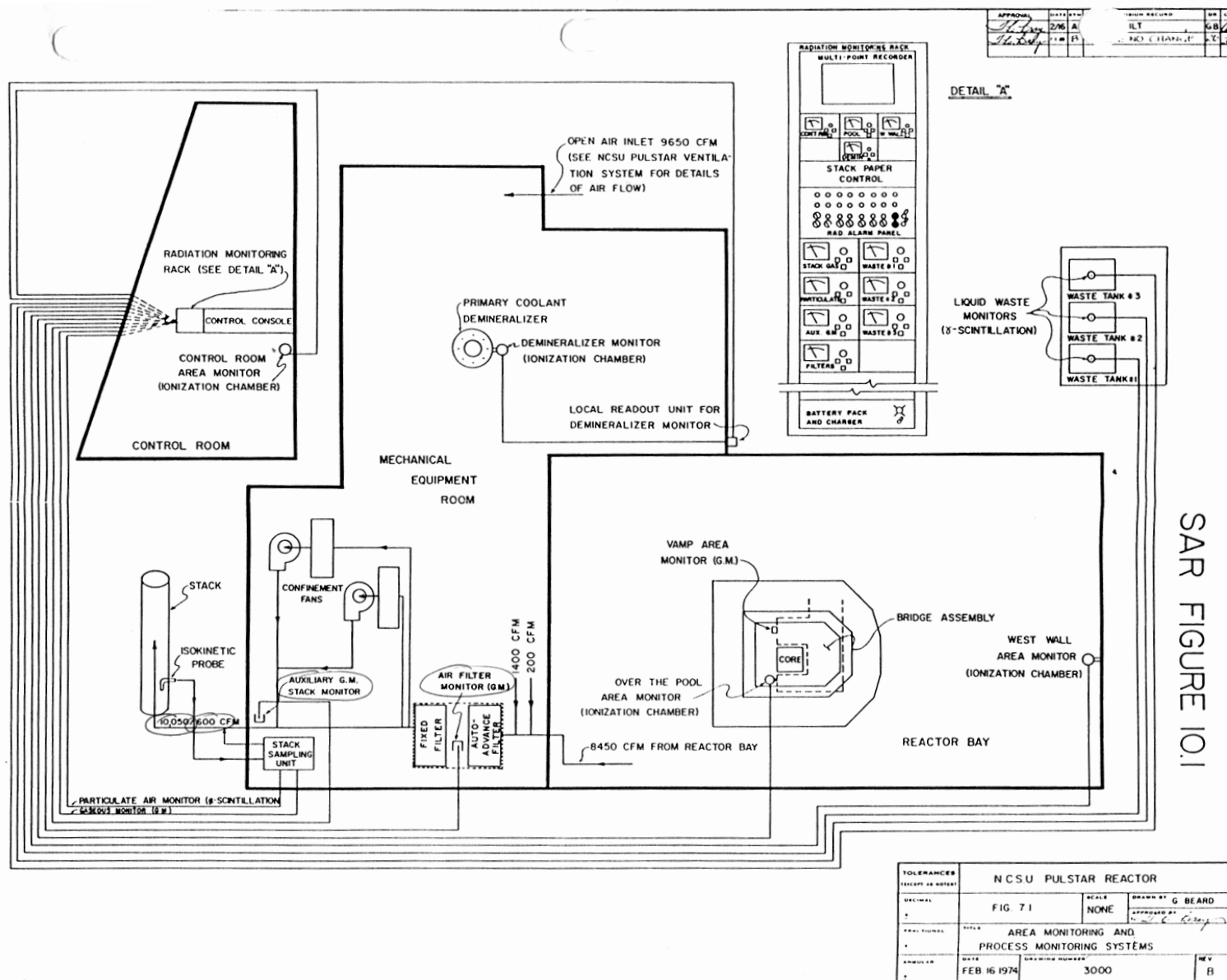


PULSTAR Confinement & HVAC Systems

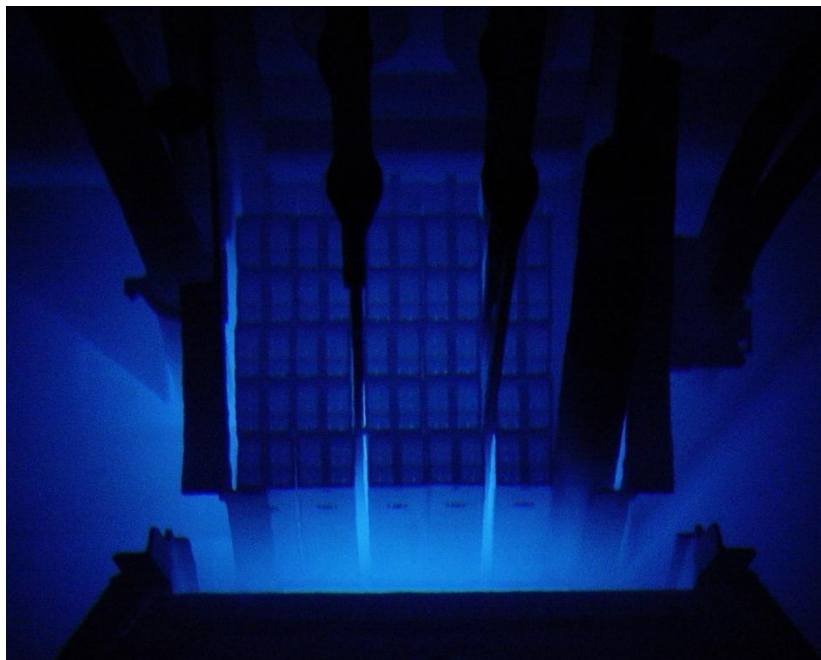


PULSTAR Radiation Monitoring and EVAC System

OPERATIONS MANUAL FIGURE 7.1



Next Week



Walkthrough Reactor:

- Over the next two weeks, we will be walking through the bay area (and cooling tower, and ventilation room, if time permits).
- Wear closed-toed shoes and long pants for lab.
- Bring a copy of photo ID
- A sign-up for the walkthrough will release on Moodle next Monday after class

Reactor Operator Training:

- Cecilia Harrison and Kellen Williamson, current Student Reactor Operators, will be starting the training lecture series after the class lecture has concluded on Tuesdays.
- Their first lecture will be on Aug 25